IBM Cognos Analytics Project

Case Study 11 : CodeVision IT Solutions - Project & Resource Analytics.

College Code : 9222

College Name: Theni Kammavar Sangam College of Technology

NM ID : E38D78D1903E3308CB4488C74C9014D2

Department : BTech (IT)

Name : Pavithra Devi S

[Reporting : 1] Create a Project Health Report showing Project Name, Client, Budget vs Actual Cost,Planned vs Actual Hours, and Cost Overrun % (calculated). Apply Conditional Styling: Cost Overrun > 20% in Red, 10–20% in Amber, < 10% in Green. Add Header with CodeVisionbranding, Footer with 'Management Confidential'. Add prompts for Client Country and Project Type.

PAGE 1 :

CodeVision IT Solutions – Project Health Report

Project Name	Client Name	Budget (USD)	Actual Cost (USD)	Planned Hours	Actual Hours
Project_40	StartupXYZ India	464182	416350.5	629	691
Project_19	RetailGiant USA	463333	558050.61	932	721
Project_15	RetailGiant USA	1165607	1059594.8	3904	3568
Project_23	StartupXYZ India	650630	635861.37	1170	1227
Project_36	HealthPlus UK	956752	1077199.09	3901	3187
Project_26	TechCorp USA	402561	320686.92	2042	2074
Project_42	StartupXYZ India	280409	233851.48	136	154
Project_16	FinBank UK	387361	278676.59	400	284
Project_17	FinBank UK	304184	234644.16	2759	2652
Project_18	RetailGiant USA	174034	138258.37	696	601
Project_34	FinBank UK	186587	138385.95	1138	808
Project_02	StartupXYZ India	340982	358814.08	878	1003
Project_22	StartupXYZ India	425254	511240.81	2214	2120
Project_15	FinBank UK	56641	45963.79	905	928
Project_31	HealthPlus UK	634463	532831.89	1876	1958
Project_01	FinBank UK	243892	307422.49	2817	2887
Project_02	FinBank UK	389741	399330.13	1820	1476
Project_30	RetailGiant USA	226772	194361.53	418	340
Project_30	StartupXYZ India	878618	1005923.74	1344	1262
Project_35	FinBank UK	526070	503541.96	2704	2581
Project_38	StartupXYZ India	319034	332007.31	1447	1646
Project_16	TechCorp USA	64564	49200.74	1048	1174
Project_02	RetailGiant USA	783084	691053.87	3678	3651
Project_25	FinBank UK	288190	300370.09	1838	2016
Project_08	RetailGiant USA	411069	534310.31	559	435
Project_38	HealthPlus UK	38117	30990.06	1316	1627
Project_37	HealthPlus UK	213000	242219.65	168	159
Project_39	RetailGiant USA	201089	249889.34	1913	2302
Project_28	HealthPlus UK	316118	268303.25	703	581
Project_20	HealthPlus UK	435996	393154.5	2363	1930

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CodeVision IT Solutions – Project Health Report

Project Name	Client Name	Budget (USD)	Actual Cost (USD)	Planned Hours	Actual Hours
Project_33	StartupXYZ India	316184	229202.47	880	1059
Project_08	FinBank UK	321762	287596.85	1402	1549
Project_14	FinBank UK	270331	315006.38	381	304
Project_37	RetailGiant USA	496514	517490.47	1759	2228
Project_46	TechCorp USA	375271	369862.4	1619	1902
Project_11	StartupXYZ India	87086	78759.03	347	403
Project_05	StartupXYZ India	225972	256886.72	937	1070
Project_33	TechCorp USA	278496	320573.56	1180	1472
Project_28	TechCorp USA	321355	323246.34	437	353
Project_39	TechCorp USA	754212	799476.67	1588	1154
Project_10	FinBank UK	227684	176789.08	567	640
Project_30	FinBank UK	807486	741537.42	2095	1861
Project_16	RetailGiant USA	164114	175853.53	3132	2781
Project_44	TechCorp USA	196905	176143.28	283	291
Project_32	StartupXYZ India	774774	805778.92	5482	5028
Project_11	RetailGiant USA	198338	150710.23	1519	1262
Project_20	StartupXYZ India	429250	350362.68	1240	1258
Project_31	RetailGiant USA	463962	474794.95	3223	4116
Project_21	RetailGiant USA	136581	102784.51	1101	1176
Project_17	StartupXYZ India	157271	151685.9	143	107
Project_44	FinBank UK	344230	356337.15	440	437
Project_11	FinBank UK	518992	573834.92	2507	2038
Project_18	StartupXYZ India	495675	520048.32	2086	2604
Project_06	StartupXYZ India	202661	206724.45	2820	3024
Project_41	FinBank UK	505666	601356.92	2551	2257
Project_41	StartupXYZ India	307819	235627.78	1025	1080
Project_24	TechCorp USA	31968	39894.41	1523	1528
Project_24	RetailGiant USA	1094951	1278332.94	2244	2051
Project_43	FinBank UK	389151	412641.56	492	639
Project_42	HealthPlus UK	322743	248733.98	3049	2603

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PAGE 3 :

CodeVision IT Solutions – Project Health Report

Project Name	Client Name	Budget (USD)	Actual Cost (USD)	Planned Hours	Actual Hours
Project_25	TechCorp USA	350290	285900.55	251	316
Project_45	TechCorp USA	511666	486137.71	2412	2083
Project_22	RetailGiant USA	278775	328393.98	1729	1489
Project_25	StartupXYZ India	38738	38070.83	311	402
Project_35	StartupXYZ India	250431	316158.91	484	460
Project_43	StartupXYZ India	329652	340313.54	1937	2402
Project_47	RetailGiant USA	441298	461009.81	1792	1589
Project_01	StartupXYZ India	145257	178678.69	774	803
Project_34	TechCorp USA	357858	463292.92	1029	1060
Project_07	HealthPlus UK	45340	46343.71	1912	2345
Project_45	StartupXYZ India	210086	212979.42	525	588
Project_29	StartupXYZ India	240642	304716.5	1825	1682
Project_27	StartupXYZ India	491306	547159.17	1194	1311
Project_23	RetailGiant USA	341109	255759.2	1253	1604
Project_19	HealthPlus UK	257983	191215.56	1128	1164
Project_06	HealthPlus UK	400623	436652.5	2799	2728
Project_23	HealthPlus UK	202874	194335.93	639	750
Project_24	StartupXYZ India	450784	471818.43	1599	1124
Project_48	HealthPlus UK	292599	271744.15	248	279
Project_44	RetailGiant USA	755309	601895.75	1548	1900
Project_01	HealthPlus UK	467788	585196.99	1671	1478
Project_03	TechCorp USA	80043	102859.65	1469	1570
Project_22	HealthPlus UK	267796	303552.61	1516	1506
Project_14	HealthPlus UK	301552	249694.35	371	393
Project_39	HealthPlus UK	287989	341773.51	2270	2854
Project_33	FinBank UK	397627	463920.85	1848	1727
Project_06	TechCorp USA	335399	266226.87	261	333
Project_43	RetailGiant USA	231368	202872.83	3132	2870
Project_36	FinBank UK	445588	339164.39	1624	1263
Project_04	TechCorp USA	437752	566511.61	1756	1642

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PAGE 4 :

CodeVision IT Solutions – Project Health Report

Project Name	Client Name	Budget (USD)	Actual Cost (USD)	Planned Hours	Actual Hours
Project_07	StartupXYZ India	280767	294967.09	3121	3464
Project_31	StartupXYZ India	613775	706264.33	2837	2795
Project_46	FinBank UK	545589	469671.95	2007	2259
Project_28	StartupXYZ India	19605	24980.63	1552	1105
Project_27	TechCorp USA	433036	356313.64	1035	1062
Project_47	HealthPlus UK	225297	197136.6	2334	2247
Project_42	TechCorp USA	470305	463135.17	2703	2851
Project_07	FinBank UK	398062	292320.64	1986	1945
Project_25	RetailGiant USA	127178	90422.65	1313	1480
Project_44	StartupXYZ India	534539	544058.95	2422	2249
Project_14	StartupXYZ India	223851	185202.59	204	158
Project_35	HealthPlus UK	761693	657458.32	2723	2668
Project_06	FinBank UK	35200	27778.37	124	133
Project_19	TechCorp USA	173533	157198.97	2128	2226
Project_33	RetailGiant USA	190095	189684.91	109	116
Project_09	FinBank UK	288854	203638.74	1519	1934
Project_09	StartupXYZ India	495082	491147.69	2010	1964
Project_29	FinBank UK	149439	161788.05	566	509
Project_34	RetailGiant USA	594578	584972.71	1995	1671
Project_40	FinBank UK	164872	185698.55	866	1109
Project_10	StartupXYZ India	313245	290662.44	1556	1160
Project_11	TechCorp USA	58361	62506.29	1947	2515
Project_27	HealthPlus UK	86215	109640.35	1898	1722
Project_29	HealthPlus UK	61519	57565.6	1557	1361
Project_15	TechCorp USA	314065	344641.87	726	744
Project_18	FinBank UK	117171	134259	1935	1423
Project_34	HealthPlus UK	335438	369394.5	555	625
Project_10	RetailGiant USA	29306	27605.18	1474	1816
Project_03	FinBank UK	398892	351310.93	1969	1712
Project_49	TechCorp USA	574806	490937.38	2764	2391

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PAGE 5 :

CodeVision IT Solutions – Project Health Report

Project Name	Client Name	Budget (USD)	Actual Cost (USD)	Planned Hours	Actual Hours
Project_32	RetailGiant USA	392324	436715.5	1621	1362
Project_05	TechCorp USA	426436	472859.63	2302	2624
Project_10	TechCorp USA	261221	331632.56	170	119
Project_24	HealthPlus UK	519677	393203.31	1489	1551
Project_17	RetailGiant USA	30422	28090.71	1936	1942
Project_15	HealthPlus UK	301148	237757.9	631	796
Project_04	HealthPlus UK	96597	99945.29	1360	1541
Project_44	HealthPlus UK	326382	278030.41	871	904
Project_39	FinBank UK	294231	282757.1	1540	1825
Project_02	TechCorp USA	195148	225883.42	570	617
Project_38	TechCorp USA	26057	28961.43	1050	739
Project_09	HealthPlus UK	143550	178732.03	587	420
Project_16	StartupXYZ India	364746	332524.16	1207	1375
Project_29	TechCorp USA	305550	259585.78	320	365
Project_26	RetailGiant USA	110090	136350.89	1293	1441
Project_07	TechCorp USA	420634	373620.6	2064	2224
Project_47	StartupXYZ India	182143	144342.73	1019	749
Project_21	HealthPlus UK	156907	188699.99	523	600
Project_21	StartupXYZ India	252577	282406.99	1371	1335
Project_08	HealthPlus UK	328652	410111.32	678	718
Project_03	RetailGiant USA	198979	220108.39	1064	1148
Project_33	HealthPlus UK	15300	16326.21	556	512
Project_50	RetailGiant USA	81304	80941.4	527	467

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[Reporting : 2] Build an Employee Attrition Report listing employees who left the organization in the current year with their Department, Experience Band, Exit Reas on, and Performance Rating.Apply Conditional Styling to highlight employees with Performance Rating > 3 who left (unexpected loss) in Red. Summarize attrition count by department in the footer. Use Segoe UI font, size 11.

PAGE 1 :

Employee Name	Department	Experience (years)	Exit Reason	Performance Rating (1-5)	Attrition Flag
Employee_003	Software Dev	11.8	Relocation	3.7	Yes
Employee_004	Support	13.7	Higher Salary	3.1	Yes
Employee_011	Software Dev	9.7	Better Opportunity	2	Yes
Employee_016	DevOps	5.1	Further Studies	2.2	Yes
Employee_020	UI/UX	3.1	Relocation	3.9	Yes
Employee_021	Support	6.6	Personal	4.1	Yes
Employee_023	Support	11.6	Further Studies	2.4	Yes
Employee_025	Software Dev	1.9	Relocation	2.5	Yes
Employee_028	Software Dev	8.9	Personal	3	Yes
Employee_030	DevOps	2.4	Higher Salary	1.7	Yes
Employee_031	Cloud	11.7	Personal	3.8	Yes
Employee_033	Software Dev	2.8	Better Opportunity	2	Yes
Employee_034	Data Analytics	10.6	Personal	3.4	Yes
Employee_039	QA	5.3	Better Opportunity	1.9	Yes
Employee_040	Cloud	10.8	Personal	1	Yes
Employee_041	QA	1.2	Further Studies	5	Yes
Employee_049	DevOps	13.1	Further Studies	5	Yes
Employee_051	QA	6.2	Higher Salary	4.8	Yes
Employee_053	Software Dev	0.6	Personal	4.3	Yes
Employee_056	UI/UX	9.3	Higher Salary	2.6	Yes
Employee_060	Software Dev	6.7	Further Studies	2.2	Yes
Employee_064	Data Analytics	8.7	Further Studies	1.3	Yes
Employee_072	QA	6.6	Further Studies	2	Yes
Employee_079	Data Analytics	5.8	Higher Salary	2.3	Yes
Employee_086	UI/UX	13.6	Higher Salary	2.8	Yes

Summarize Attritution Count

PAGE 2 :

Employee Name	Department	Experience (years)	Exit Reason	Performance Rating (1-5)	Attrition Flag
Employee_093	Cloud	5.8	Better Opportunity	2.6	Yes
Employee_096	DevOps	8.9	Further Studies	3	Yes
Employee_097	Data Analytics	6.4	Better Opportunity	1.1	Yes
Employee_101	Cloud	12.8	Further Studies	4.5	Yes
Employee_106	Software Dev	8.1	Personal	3.8	Yes
Employee_109	UI/UX	9.3	Further Studies	4.7	Yes
Employee_111	Software Dev	9.9	Better Opportunity	1.1	Yes
Employee_114	QA	12.6	Better Opportunity	2.1	Yes
Employee_118	UI/UX	4.2	Further Studies	4.6	Yes
Employee_130	UI/UX	5.3	Personal	3.1	Yes
Employee_147	Data Analytics	12.6	Better Opportunity	4.2	Yes
Employee_149	Software Dev	8.9	Better Opportunity	1.7	Yes
Employee_157	UI/UX	13	Further Studies	2.6	Yes
Employee_158	Software Dev	14.7	Personal	2.7	Yes
Employee_159	DevOps	4.1	Better Opportunity	3	Yes
Employee_162	Support	5.1	Further Studies	4.9	Yes
Employee_170	Software Dev	8.3	Better Opportunity	4.9	Yes
Employee_175	Cloud	13.7	Better Opportunity	2.8	Yes
Employee_176	UI/UX	11.2	Higher Salary	3.8	Yes
Employee_178	Software Dev	4.5	Better Opportunity	4.3	Yes
Employee_181	Support	8.6	Personal	1.6	Yes
Employee_195	Cloud	8.3	Relocation	2.3	Yes
Employee_200	Support	8.8	Relocation	1.8	Yes
Overall - Summary					

Summarize Attritution Count

PAGE 3 :

Employee Name	Department	Experience (years)	Exit Reason	Performance Rating (1-5)	Attrition Flag
Overall - Count	**48**				

Summarize Attritution Count

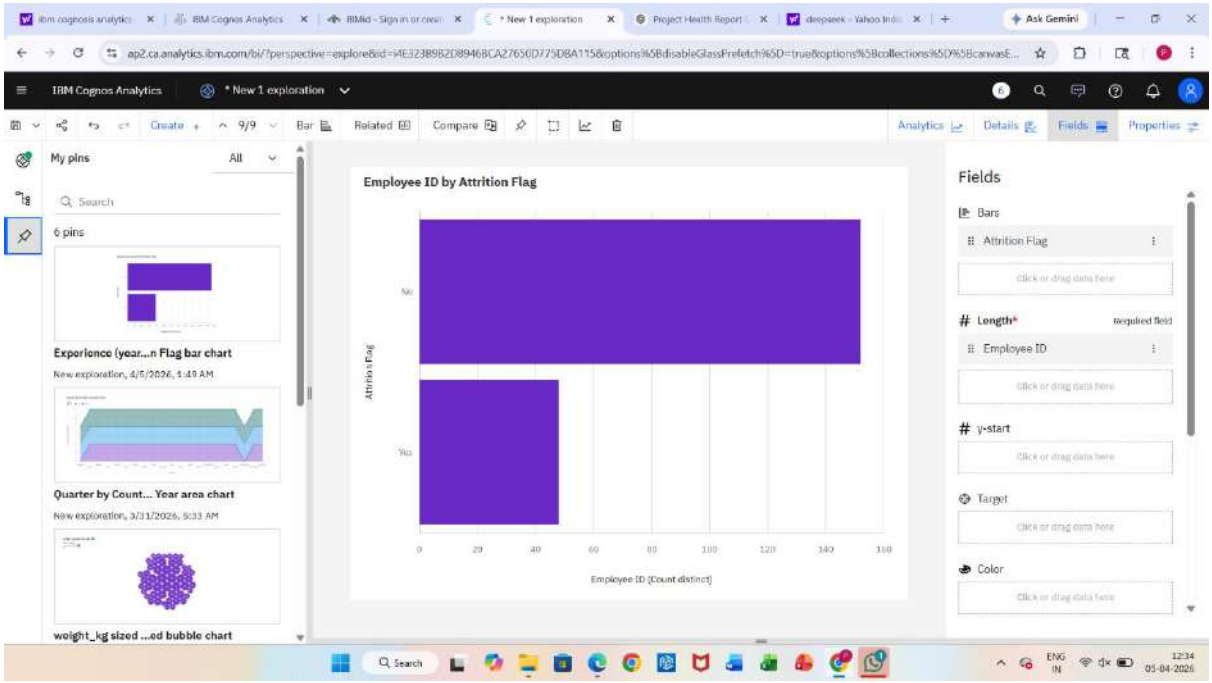
[Reporting : 3] Using Report Visualize Prompt, allow HR and Finance Managers to select chart type to view Monthly Billable vs Non-Billable Hours comparison across departments. Add prompts for Department and Technology Stack. Summarize overall Billability Rate % and highlight departments below 75% billability using Conditional Styling in Amber.

PAGE 1 :

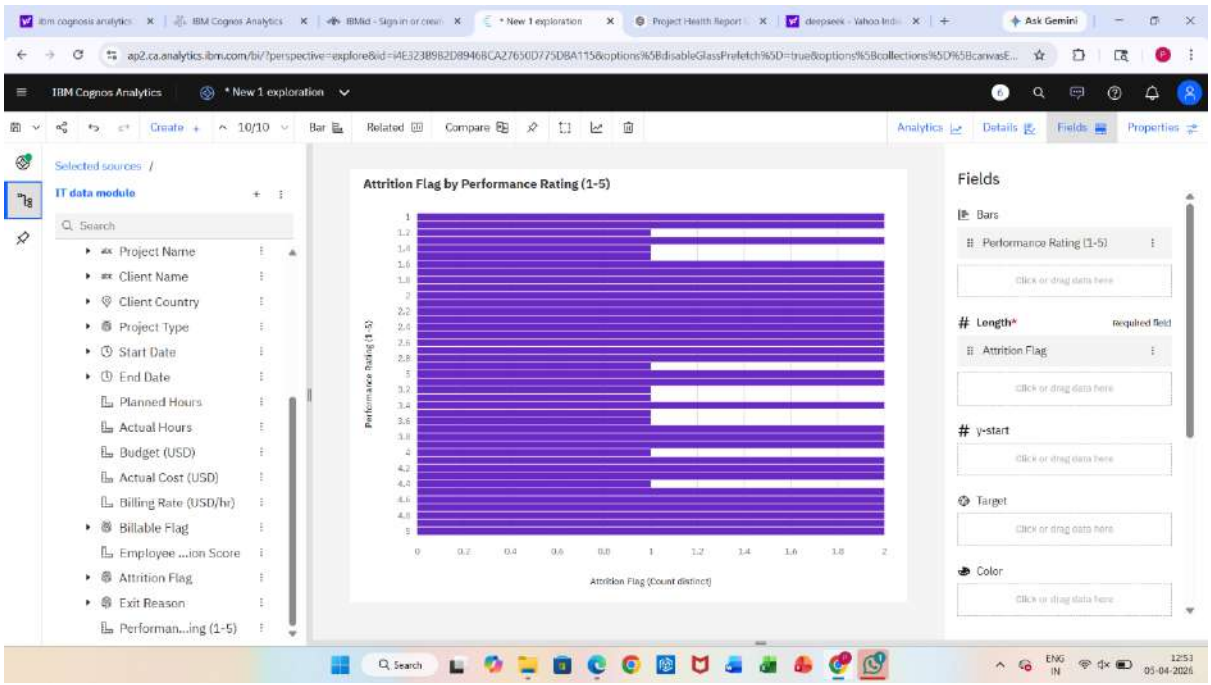
PAGE 2 :

[Exploration : 4] Use Cognos Exploration to investigate what factors contribute to employee attrition at CodeVision. Explore relationships between Experience Band, Performance Rating, and Attrition. Also explore which Project Types (Fixed vs T&M) are more likely to exceed budget. Document top 3 attrition risk factors and top 3 project overrun predictors.

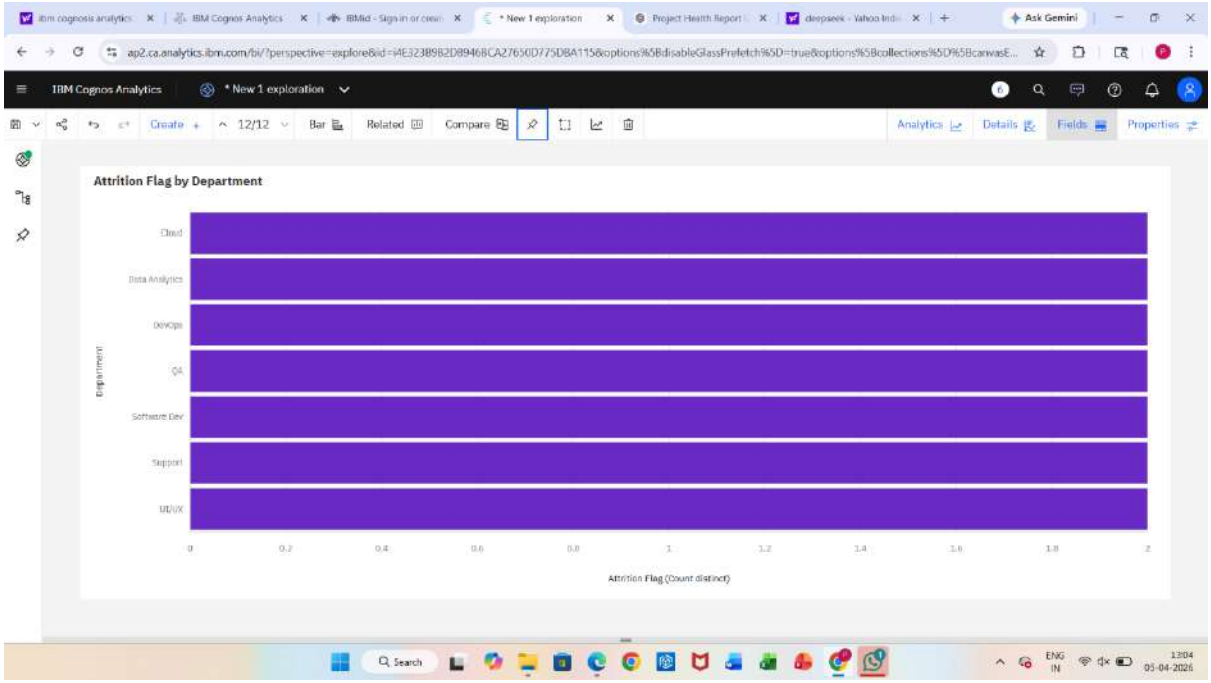
PAGE 1 :



PAGE 2 :



PAGE 3 :


IBM Cognos Analytics
* New 1 exploration
Create
12/12
Bar
Related
Compare
Analytics
Details
Fields
Properties
Attrition Flag by Department
Cloud
Data Analytics
DevOps
QA
Software Dev
Support
UI/UX
Department
0
0.2
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0.6
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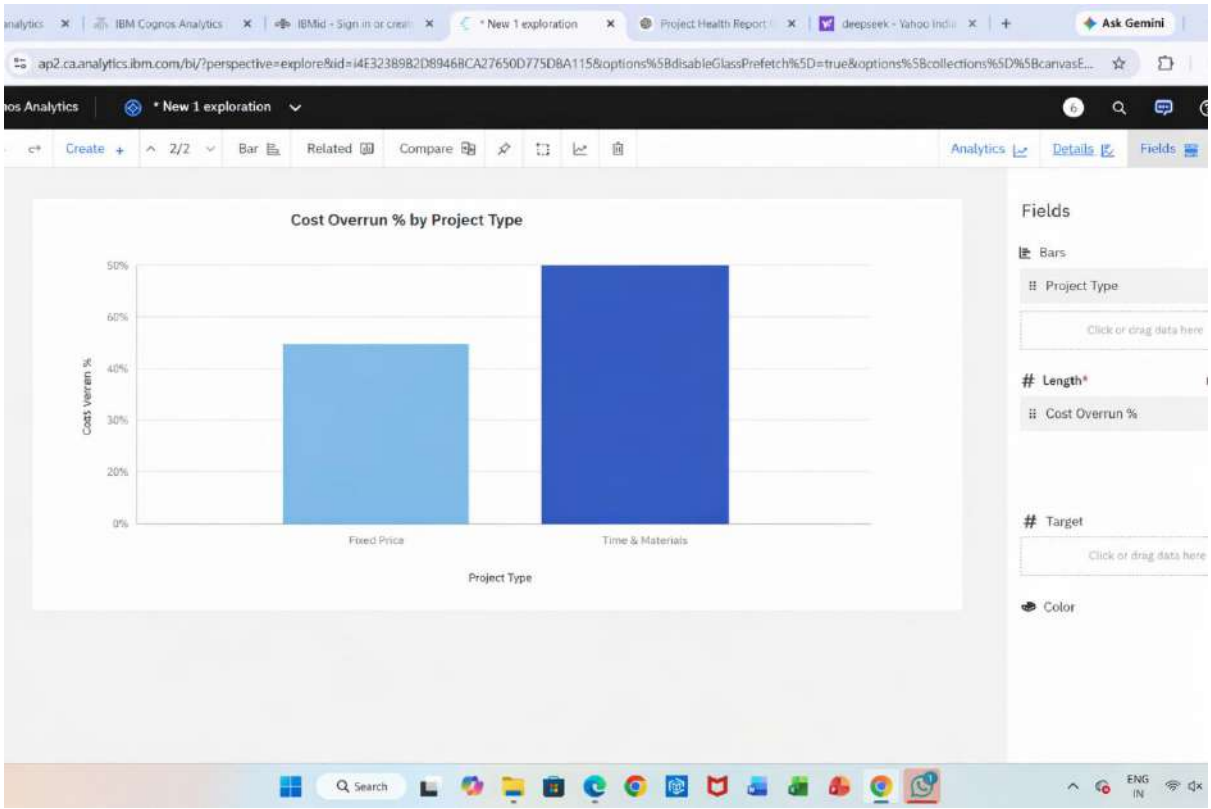
1
1.2
1.4
1.6
1.8
2
Attrition Flag (Count distinct)

PAGE 4 :


IBM Cognos Analytics
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Create
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Related
Compare
Analytics
Details
Fields
Cost Overrun % by Project Type
50%
60%
40%
30%
20%
0%
Cost Overrun %
Fixed Price
Time & Materials
Project Type
Fields
Bars
Project Type
Click or drag data here
Length*
Cost Overrun %
Target
Click or drag data here
Color

Top 3 Attrition Risk Factors :

1. **Low Experience Employees**
   - Employees with less experience (0–2 years) have higher attrition
2. **High-Performing Employees Leaving**
   - High-rated employees are also leaving, leading to talent loss
3. **Department-wise Attrition**
   - Certain departments (such as IT or Sales) show higher attrition

Top 3 Project Overrun Predictors :

1. **Fixed Projects**
   - Fixed-price projects are more likely to exceed budget
2. **Poor Planning and Estimation**
   - Inaccurate planning leads to cost overruns
3. **Resource Allocation Issues**
   - Lack of proper resources increases project cost

[Dashboard : 5] Design an IT Operations Dashboard with: (1) KPI tiles — Active Projects, Overall Billability Rate, Attrition Rate %, Average Employee Satisfaction Score, (2) A Bar chart for Projects by Client Country, (3) A Line chart for Monthly Billable Hours Trend, (4) A Pie chart for Project Type Distribution, (5) An At-Risk Projects table (Overrun > 10%). Add Department and Client filters.

CREATING A PROFESSIONAL WEB APP FOR DASHBOARD :

HTML CODE :

```
<!DOCTYPE html>
<html lang="ta">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CodeVision Analytics</title>
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&dis
play=swap" rel="stylesheet">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.0.0-beta3/css/all.min.css">
  <script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.min.js"></script>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Inter', sans-serif;
```

```
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    min-height: 100vh;
}
/* Login Page */
.login-container {
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}
.login-card {
    background: white;
    border-radius: 32px;
    padding: 40px;
    width: 400px;
    box-shadow: 0 25px 50px rgba(0,0,0,0.2);
}
.login-card h1 {
    text-align: center;
    margin-bottom: 30px;
    color: #1e293b;
}
.input-group {
    margin-bottom: 20px;
}
.input-group label {
    display: block;
    margin-bottom: 8px;
```

```
    font-weight: 600;
    color: #334155;
}
.input-group input {
    width: 100%;
    padding: 12px 16px;
    border: 2px solid #e2e8f0;
    border-radius: 12px;
    font-size: 14px;
}
.login-btn {
    width: 100%;
    background: linear-gradient(135deg, #667eea, #764ba2);
    color: white;
    border: none;
    padding: 14px;
    border-radius: 12px;
    font-weight: 600;
    cursor: pointer;
}
.error-msg {
    color: red;
    text-align: center;
    margin-top: 15px;
}
/* Dashboard Page */
.dashboard-container {
    display: none;
```

```
  background: #f1f5f9;
  min-height: 100vh;
}
.navbar {
  background: white;
  padding: 15px 30px;
  display: flex;
  justify-content: space-between;
  align-items: center;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}
.logout-btn {
  background: #ef4444;
  color: white;
  border: none;
  padding: 8px 20px;
  border-radius: 8px;
  cursor: pointer;
}
.kpi-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 20px;
  padding: 30px;
}
.kpi-card {
  background: white;
  padding: 20px;
```

```
    border-radius: 16px;
    text-align: center;
    box-shadow: 0 2px 5px rgba(0,0,0,0.05);
}
.kpi-value {
    font-size: 32px;
    font-weight: bold;
    color: #667eea;
    margin: 10px 0;
}
.charts-row {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(400px, 1fr));
    gap: 20px;
    padding: 0 30px 30px 30px;
}
.chart-card {
    background: white;
    padding: 20px;
    border-radius: 16px;
}
.table-section {
    background: white;
    margin: 0 30px 30px 30px;
    border-radius: 16px;
    padding: 20px;
    overflow-x: auto;
}
```

```
      table {
        width: 100%;
        border-collapse: collapse;
      }
      th, td {
        padding: 12px;
        text-align: left;
        border-bottom: 1px solid #e2e8f0;
      }
      th {
        background: #f8fafc;
      }
      iframe {
        width: 100%;
        height: 400px;
        border: none;
        border-radius: 16px;
      }
      .iframe-container {
        margin: 0 30px 30px 30px;
        background: white;
        border-radius: 16px;
        padding: 20px;
      }
    </style>
  </head>
  <body>
```

```
<!-- LOGIN PAGE -->
<div id="loginPage" class="login-container">
  <div class="login-card">
    <h1><i class="fas fa-chart-line"></i> CodeVision</h1>
    <div class="input-group">
      <label>Email</label>
      <input type="email" id="email" placeholder="demo@codevision.com"
value="demo@codevision.com">
    </div>
    <div class="input-group">
      <label>Password</label>
      <input type="password" id="password" placeholder="Enter password"
value="admin123">
    </div>
    <button class="login-btn" onclick="login()">Login</button>
    <div id="loginError" class="error-msg"></div>
    <p style="text-align: center; margin-top: 20px; font-size: 12px;">Demo:
admin123</p>
  </div>
</div>

<!-- DASHBOARD PAGE -->
<div id="dashboardPage" class="dashboard-container">
  <div class="navbar">
    <h2><i class="fas fa-code"></i> CodeVision Analytics</h2>
    <button class="logout-btn" onclick="logout()">Logout</button>
  </div>

  <div class="kpi-grid" id="kpiContainer"></div>
```

```
  <div class="charts-row">
    <div class="chart-card">
      <h3>Attrition by Department</h3>
      <canvas id="deptChart"></canvas>
    </div>
    <div class="chart-card">
      <h3>Performance by Designation</h3>
      <canvas id="perfChart"></canvas>
    </div>
  </div>

  <div class="iframe-container">
    <h3>IBM Cognos Dashboard</h3>
    <iframe
src="https://ap2.ca.analytics.ibm.com/bi/?perspective=dashboard&amp;pathRef=.my
_folders%2Fquery%2B5%2Bdashboard&amp;closeWindowOnLastView=true&amp;ui_
appbar=false&amp;ui_navbar=false&amp;shareMode=embedded&amp;action=view&a
mp;mode=dashboard&amp;subView=model0000019d5ddfc32c_00000004&amp;nav_f
ilter=true"></iframe>
  </div>

  <div class="table-section">
    <h3>Employee List</h3>
    <table id="employeeTable">
      <thead>

<tr><th>ID</th><th>Name</th><th>Department</th><th>Designation</th><th>Perfor
mance</th><th>Attrition</th></tr>
      </thead>
```

```
        <tbody id="tableBody"></tbody>
      </table>
    </div>
  </div>


<script>
  // Employee Data (from your Excel file)
  const employees = [
    {id:"EMP0001", name:"Employee_001", dept:"UI/UX", desig:"Junior Dev", perf:4.5, attrition:"No", satisfaction:4.8, billable:"No"},
    {id:"EMP0002", name:"Employee_002", dept:"Cloud", desig:"Tech Lead", perf:3.8, attrition:"No", satisfaction:4.9, billable:"No"},
    {id:"EMP0003", name:"Employee_003", dept:"Software Dev", desig:"Consultant", perf:3.7, attrition:"Yes", satisfaction:1.6, billable:"Yes"},
    {id:"EMP0004", name:"Employee_004", dept:"Support", desig:"Senior Dev", perf:3.1, attrition:"Yes", satisfaction:1.5, billable:"Yes"},
    {id:"EMP0005", name:"Employee_005", dept:"Support", desig:"Junior Dev", perf:4.6, attrition:"No", satisfaction:2.9, billable:"No"},
    {id:"EMP0006", name:"Employee_006", dept:"DevOps", desig:"Analyst", perf:1.0, attrition:"No", satisfaction:3.9, billable:"No"},
    {id:"EMP0007", name:"Employee_007", dept:"QA", desig:"Tech Lead", perf:2.8, attrition:"No", satisfaction:3.0, billable:"Yes"},
    {id:"EMP0008", name:"Employee_008", dept:"DevOps", desig:"Senior Dev", perf:4.8, attrition:"No", satisfaction:4.2, billable:"No"},
    {id:"EMP0009", name:"Employee_009", dept:"QA", desig:"Manager", perf:3.1, attrition:"No", satisfaction:3.2, billable:"Yes"},
    {id:"EMP0010", name:"Employee_010", dept:"Support", desig:"Senior Dev", perf:4.9, attrition:"No", satisfaction:2.7, billable:"Yes"},
    {id:"EMP0011", name:"Employee_011", dept:"Software Dev", desig:"Manager", perf:2.0, attrition:"Yes", satisfaction:2.4, billable:"Yes"},
    {id:"EMP0012", name:"Employee_012", dept:"QA", desig:"Senior Dev", perf:4.8, attrition:"No", satisfaction:2.1, billable:"Yes"},
```

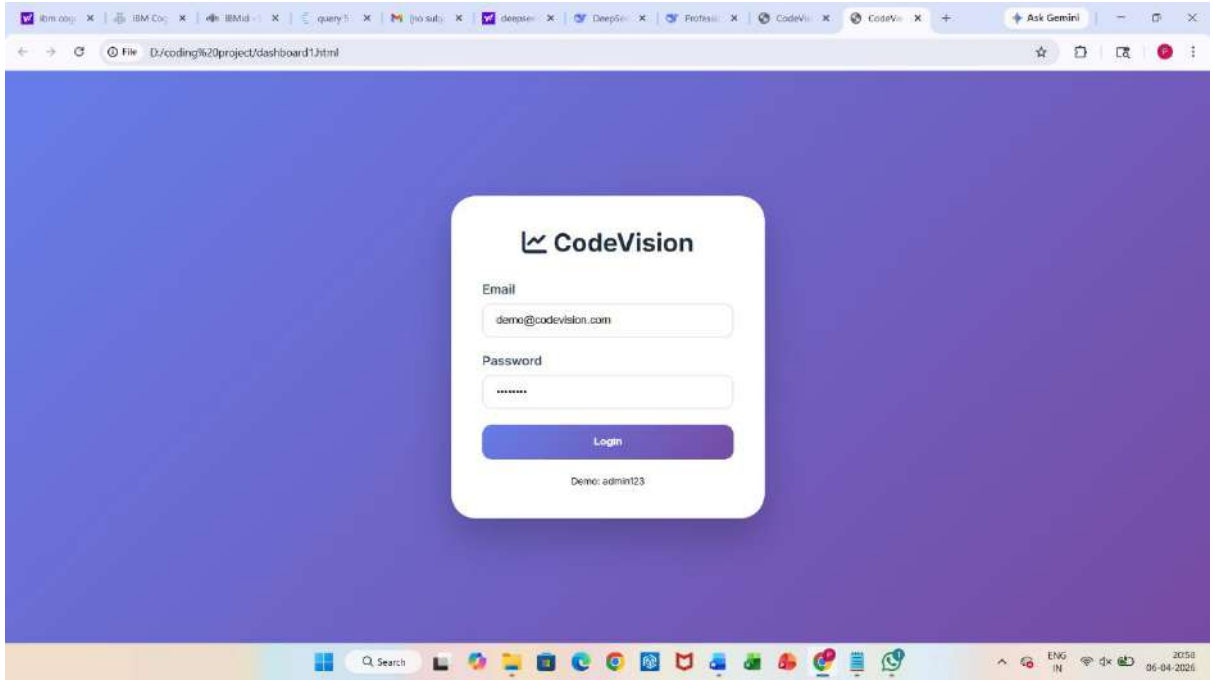
```
    {id:"EMP0013", name:"Employee_013", dept:"Data Analytics", desig:"Architect", perf:2.3, attrition:"No", satisfaction:3.0, billable:"Yes"},
    {id:"EMP0014", name:"Employee_014", dept:"UI/UX", desig:"Manager", perf:3.9, attrition:"No", satisfaction:2.9, billable:"Yes"},
    {id:"EMP0015", name:"Employee_015", dept:"UI/UX", desig:"Manager", perf:1.1, attrition:"No", satisfaction:4.1, billable:"No"},
    {id:"EMP0016", name:"Employee_016", dept:"DevOps", desig:"Consultant", perf:2.2, attrition:"Yes", satisfaction:1.2, billable:"Yes"},
    {id:"EMP0017", name:"Employee_017", dept:"Software Dev", desig:"Junior Dev", perf:1.2, attrition:"No", satisfaction:4.1, billable:"Yes"},
    {id:"EMP0018", name:"Employee_018", dept:"UI/UX", desig:"Junior Dev", perf:3.4, attrition:"No", satisfaction:1.2, billable:"Yes"},
    {id:"EMP0019", name:"Employee_019", dept:"UI/UX", desig:"Analyst", perf:1.2, attrition:"No", satisfaction:2.4, billable:"No"},
    {id:"EMP0020", name:"Employee_020", dept:"UI/UX", desig:"Consultant", perf:3.9, attrition:"Yes", satisfaction:4.9, billable:"Yes"}
  ];

  let deptChart, perfChart;

  function login() {
    const email = document.getElementById('email').value;
    const password = document.getElementById('password').value;

    if(password === 'admin123') {
      document.getElementById('loginPage').style.display = 'none';
      document.getElementById('dashboardPage').style.display = 'block';
      loadDashboard();
    } else {
      document.getElementById('loginError').innerHTML = 'Invalid password! Use: admin123';
```

```
    }
  }

  function logout() {
    document.getElementById('loginPage').style.display = 'flex';
    document.getElementById('dashboardPage').style.display = 'none';
    document.getElementById('email').value = 'demo@codevision.com';
    document.getElementById('password').value = '';
  }

  function loadDashboard() {
    // Calculate KPIs
    const total = employees.length;
    const attritionCount = employees.filter(e => e.attrition === 'Yes').length;
    const attritionRate = ((attritionCount/total)*100).toFixed(1);
    const avgPerf = (employees.reduce((sum,e) => sum + e.perf, 0)/total).toFixed(1);
    const avgSat = (employees.reduce((sum,e) => sum + e.satisfaction,
0)/total).toFixed(1);
    const billableCount = employees.filter(e => e.billable === 'Yes').length;
    const billablePct = ((billableCount/total)*100).toFixed(0);

    // Display KPI Cards
    document.getElementById('kpiContainer').innerHTML = `
      <div class="kpi-card"><h3>Total Employees</h3><div class="kpi-
value">${total}</div></div>
      <div class="kpi-card"><h3>Attrition Rate</h3><div class="kpi-
value">${attritionRate}%</div></div>
      <div class="kpi-card"><h3>Avg Performance</h3><div class="kpi-
value">${avgPerf}/5</div></div>
```

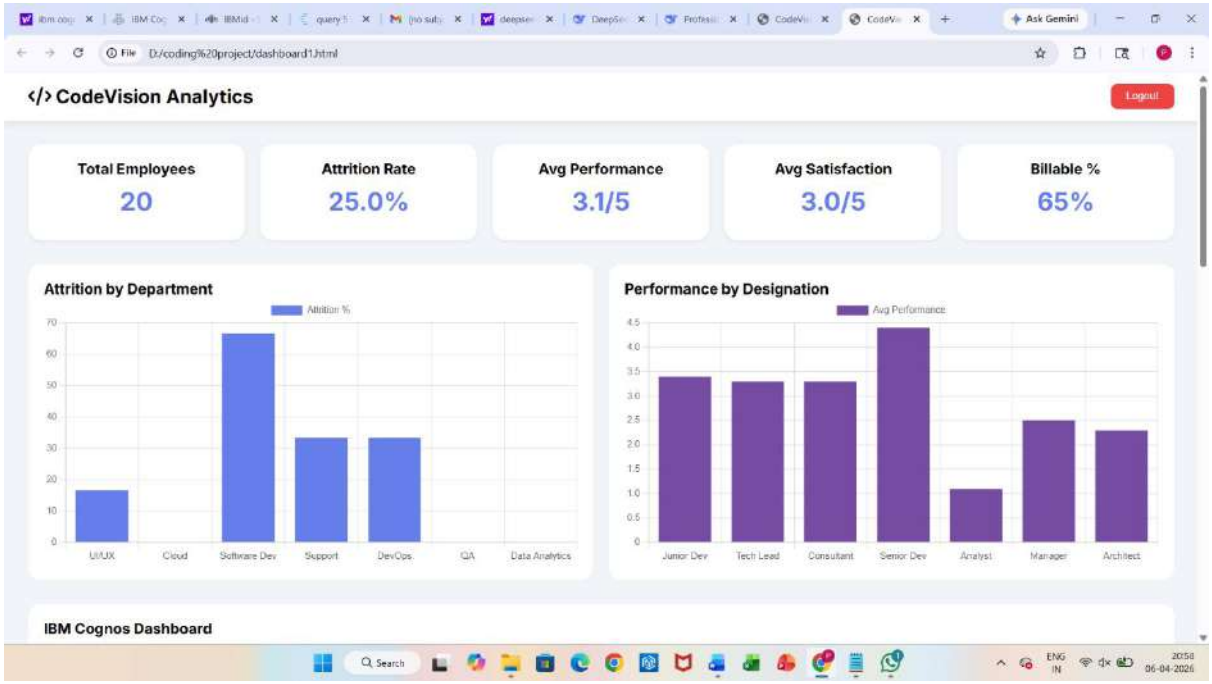
```
        <div class="kpi-card"><h3>Avg Satisfaction</h3><div class="kpi-value">${avgSat}/5</div></div>
        <div class="kpi-card"><h3>Billable %</h3><div class="kpi-value">${billablePct}%</div></div>
    `;

    // Department wise attrition
    const depts = [...new Set(employees.map(e => e.dept))];
    const deptAttrition = depts.map(dept => {
        const totalDept = employees.filter(e => e.dept === dept).length;
        const attritionDept = employees.filter(e => e.dept === dept && e.attrition === 'Yes').length;
        return (attritionDept/totalDept)*100;
    });

    if(deptChart) deptChart.destroy();
    deptChart = new Chart(document.getElementById('deptChart'), {
        type: 'bar',
        data: { labels: depts, datasets: [{ label: 'Attrition %', data: deptAttrition, backgroundColor: '#667eea' }] }
    });
    // Performance by Designation
    const desigs = [...new Set(employees.map(e => e.desig))];
    const perfByDesig = desigs.map(desig => {
        const empList = employees.filter(e => e.desig === desig);
        return (empList.reduce((sum,e) => sum + e.perf, 0)/empList.length).toFixed(1);
    });

    if(perfChart) perfChart.destroy();
```

```
    perfChart = new Chart(document.getElementById('perfChart'), {
      type: 'bar',
      data: { labels: desigs, datasets: [{ label: 'Avg Performance', data: perfByDesig,
backgroundColor: '#764ba2' }] }
    });

    // Load Table
    document.getElementById('tableBody').innerHTML = employees.map(e => `

<tr><td>${e.id}</td><td>${e.name}</td><td>${e.dept}</td><td>${e.desig}</td><td>${e.
perf}</td><td><span
style="color:${e.attrition=='Yes'?'red':'green'}">${e.attrition}</span></td></tr>
    `).join('');
  }
</script>
</body>
</html>
```

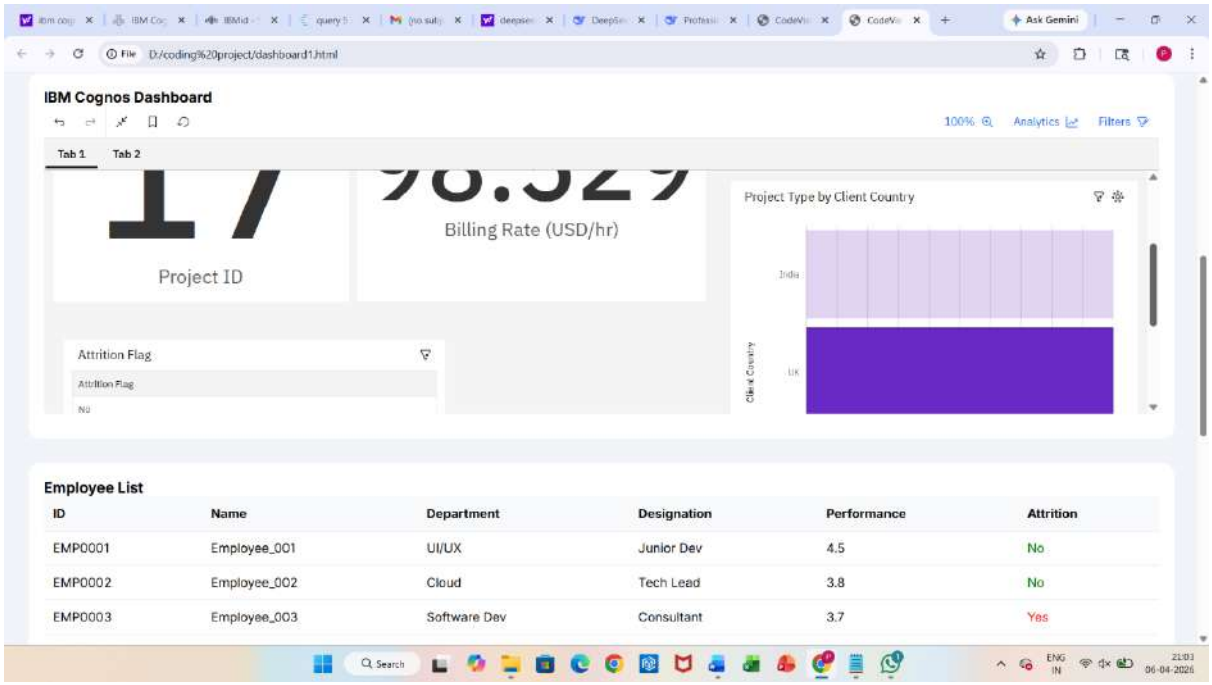
LOGIN PAGE :



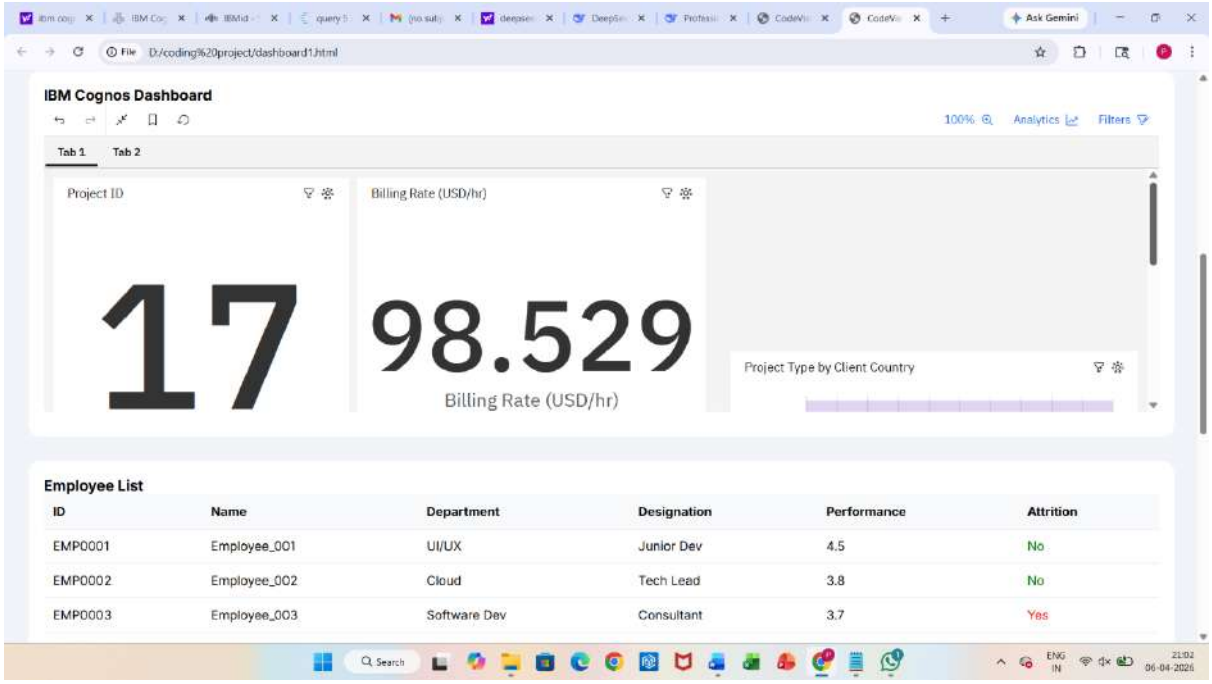
PAGE 1 :


D:/coding%20project/dashboard1.html
</> CodeVision Analytics
Logout
Total Employees
20
Attrition Rate
25.0%
Avg Performance
3.1/5
Avg Satisfaction
3.0/5
Billable %
65%
Attrition by Department
Attrition %
UI/UX
Cloud
Software Dev
Support
DevOps
QA
Data Analytics
Performance by Designation
Avg Performance
Junior Dev
Tech Lead
Consultant
Senior Dev
Analyst
Manager
Architect
IBM Cognos Dashboard

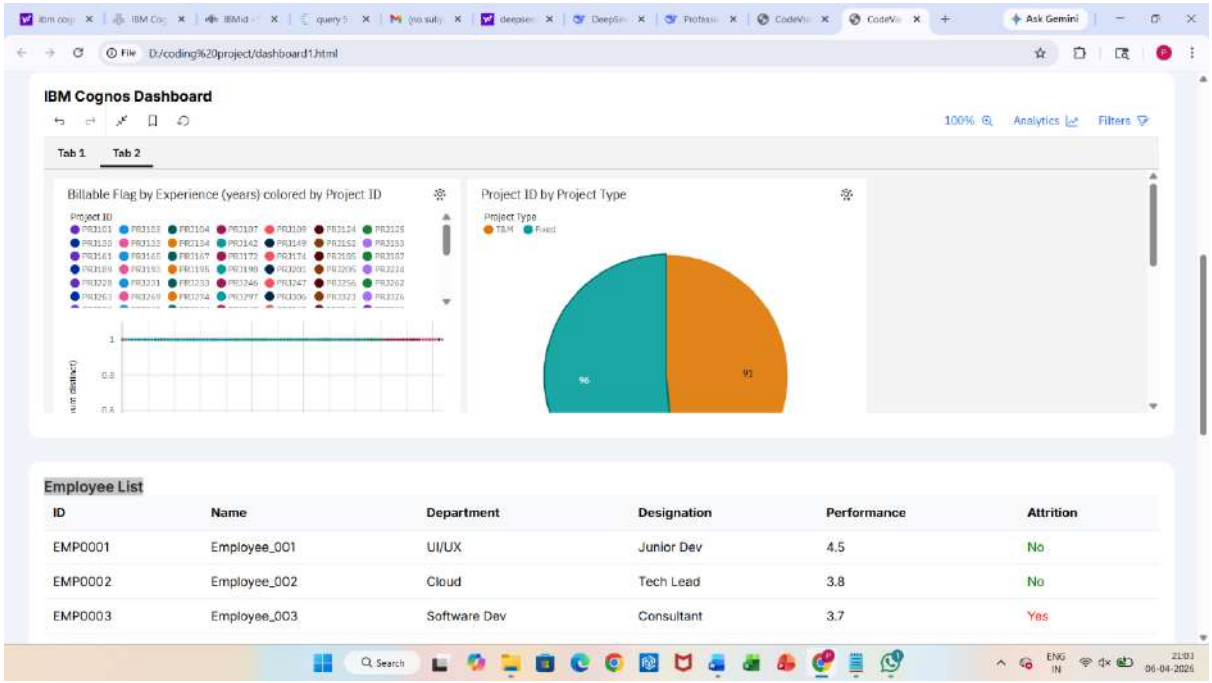
PAGE : 2


D:/coding%20project/dashboard1.html
IBM Cognos Dashboard
Tab 1
Tab 2
Analytics
Filters
Project ID
Billing Rate (USD/hr)
Project Type by Client Country
India
UK
Client Country
Attrition Flag
Employee List
ID
Name
Department
Designation
Performance
Attrition
EMP0001
Employee_001
UI/UX
Junior Dev
4.5
No
EMP0002
Employee_002
Cloud
Tech Lead
3.8
No
EMP0003
Employee_003
Software Dev
Consultant
3.7
Yes

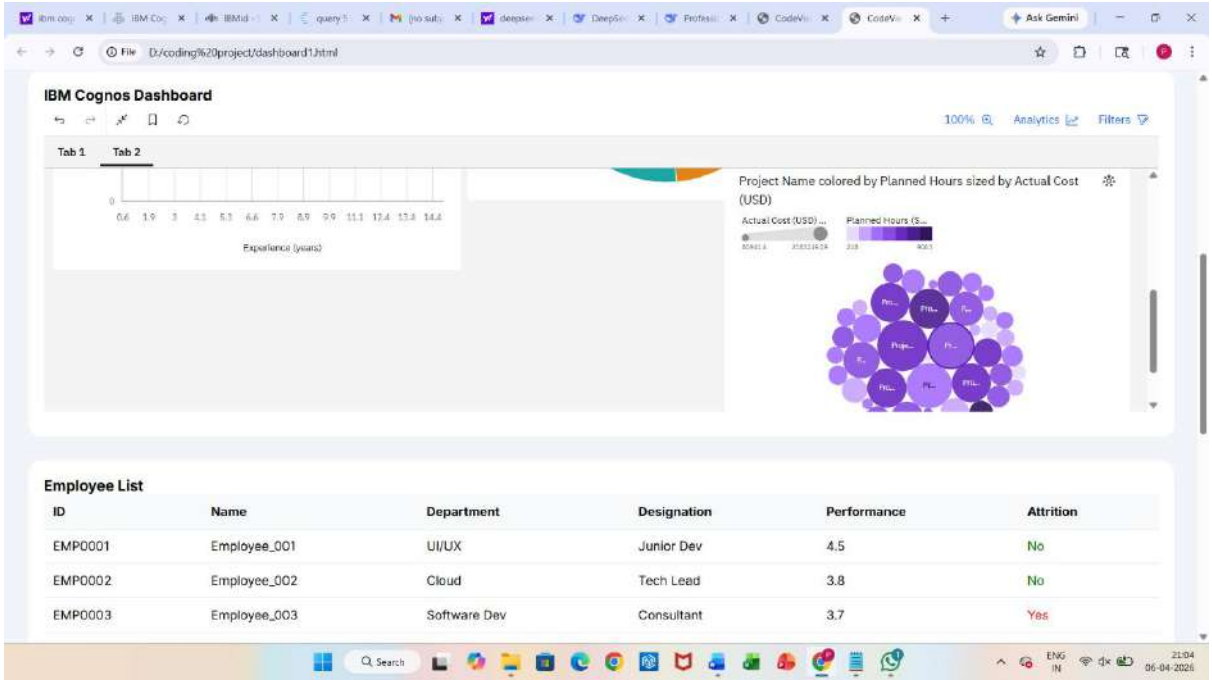
PAGE : 3


IBM Cognos Dashboard
Tab 1
Tab 2
Project ID
17
Billing Rate (USD/hr)
98.529
Billing Rate (USD/hr)
Project Type by Client Country
Employee List
ID | Name | Department | Designation | Performance | Attrition
EMP0001 | Employee_001 | UI/UX | Junior Dev | 4.5 | No
EMP0002 | Employee_002 | Cloud | Tech Lead | 3.8 | No
EMP0003 | Employee_003 | Software Dev | Consultant | 3.7 | Yes

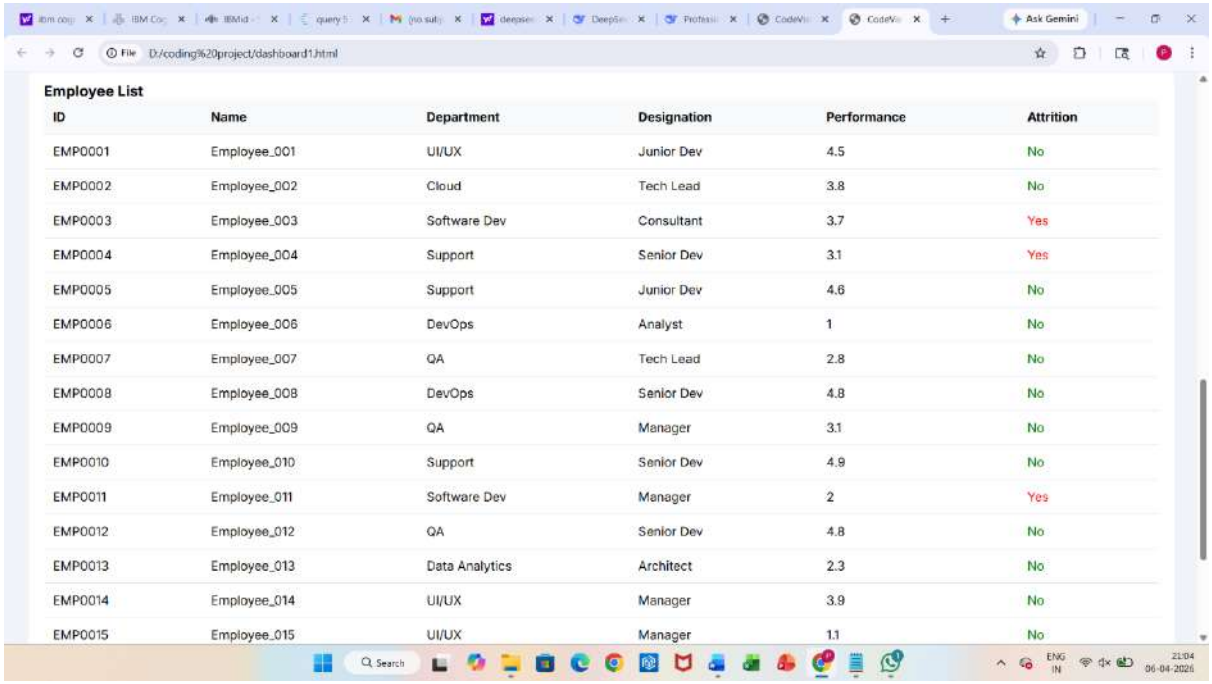
PAGE : 4


IBM Cognos Dashboard
Tab 1
Tab 2
Billable Flag by Experience (years) colored by Project ID
Project ID by Project Type
Employee List
ID | Name | Department | Designation | Performance | Attrition
EMP0001 | Employee_001 | UI/UX | Junior Dev | 4.5 | No
EMP0002 | Employee_002 | Cloud | Tech Lead | 3.8 | No
EMP0003 | Employee_003 | Software Dev | Consultant | 3.7 | Yes

PAGE : 5

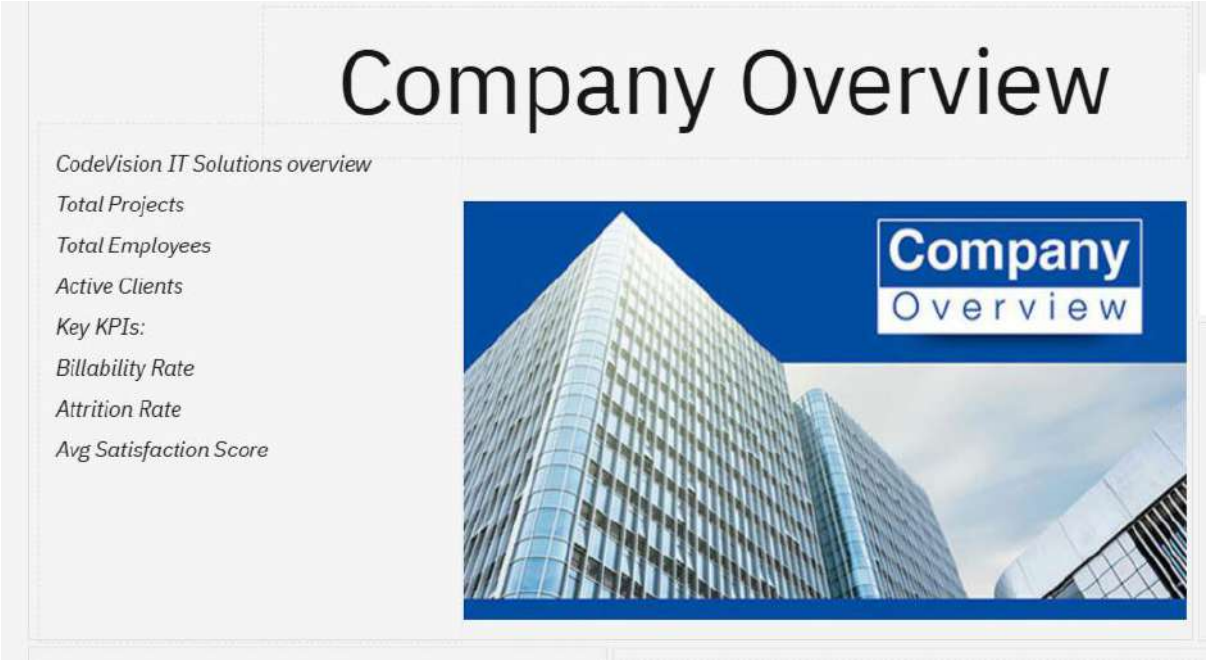

D:/coding%20project/dashboard1.html
IBM Cognos Dashboard
100%
Analytics
Filters
Tab 1
Tab 2
Experience (years)
Project Name colored by Planned Hours sized by Actual Cost (USD)
Actual Cost (USD) ...
Planned Hours (S...
Employee List
ID | Name | Department | Designation | Performance | Attrition
EMP0001 | Employee_001 | UI/UX | Junior Dev | 4.5 | No
EMP0002 | Employee_002 | Cloud | Tech Lead | 3.8 | No
EMP0003 | Employee_003 | Software Dev | Consultant | 3.7 | Yes

PAGE : 6

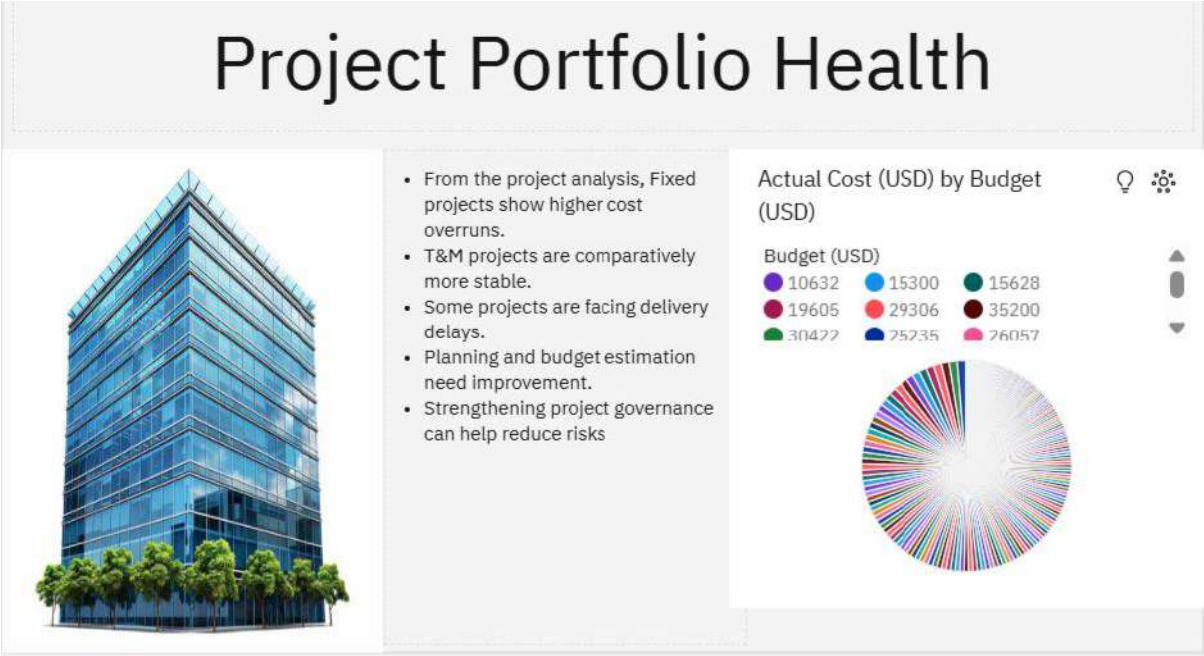

D:/coding%20project/dashboard1.html
Employee List
ID | Name | Department | Designation | Performance | Attrition
EMP0001 | Employee_001 | UI/UX | Junior Dev | 4.5 | No
EMP0002 | Employee_002 | Cloud | Tech Lead | 3.8 | No
EMP0003 | Employee_003 | Software Dev | Consultant | 3.7 | Yes
EMP0004 | Employee_004 | Support | Senior Dev | 3.1 | Yes
EMP0005 | Employee_005 | Support | Junior Dev | 4.6 | No
EMP0006 | Employee_006 | DevOps | Analyst | 1 | No
EMP0007 | Employee_007 | QA | Tech Lead | 2.8 | No
EMP0008 | Employee_008 | DevOps | Senior Dev | 4.8 | No
EMP0009 | Employee_009 | QA | Manager | 3.1 | No
EMP0010 | Employee_010 | Support | Senior Dev | 4.9 | No
EMP0011 | Employee_011 | Software Dev | Manager | 2 | Yes
EMP0012 | Employee_012 | QA | Senior Dev | 4.8 | No
EMP0013 | Employee_013 | Data Analytics | Architect | 2.3 | No
EMP0014 | Employee_014 | UI/UX | Manager | 3.9 | No
EMP0015 | Employee_015 | UI/UX | Manager | 1.1 | No

[Story Making : 6] Create a Cognos Story for CodeVision's Quarterly Leadership Review: Scene 1 — Company Performance Overview, Scene 2 — Project Portfolio Health & Delivery Status, Scene 3 — Resource Utilization & Billability Analysis, Scene 4 — Employee Attrition Trends & Exit Analysis, Scene 5 — Client-wise Revenue & Profitability, Scene 6 — Strategic Recommendations on Retention & Project Governance.

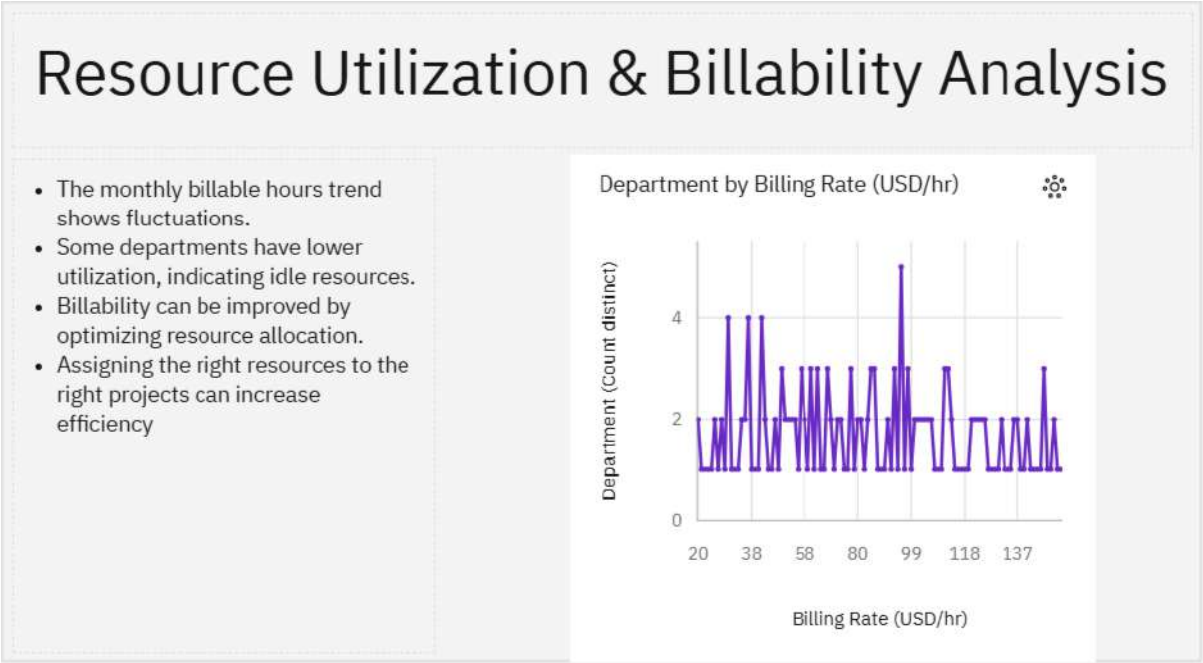
SCENE 1 :



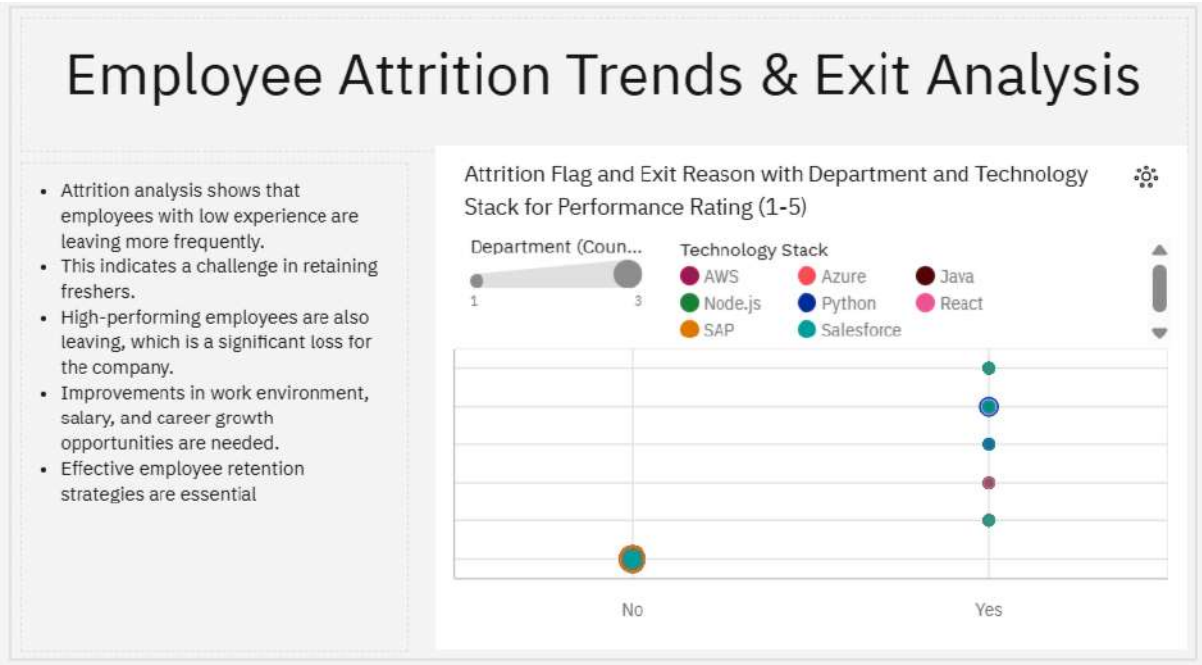
SCENE 2 :



SCENE 3 :



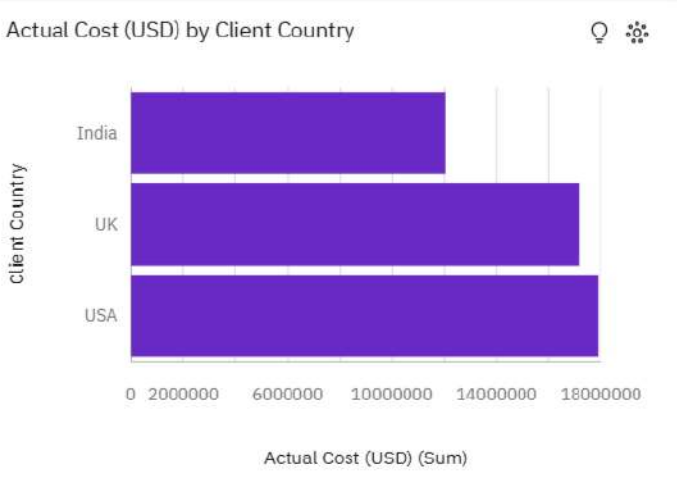
SCENE 4 :



SCENE 5 :

Client-wise Revenue & Profitability

- Client-wise analysis shows that a few key clients generate strong revenue.
- However, profitability is not consistent across all clients.
- Some projects incur higher costs, reducing profit margins.
- Focusing on high-value clients can improve overall revenue and profitability.
- Cost control and efficient delivery are important."



SCENE 6 :

Strategic Recommendations

- Implement employee retention programs
- Provide training and growth opportunities for freshers
- Improve planning and budget estimation for fixed projects
- Optimize resource utilization
- Focus on high-value clients
- Strengthen project monitoring and governance